

Development of an Artificial Intelligence-Enabled System for Smart Crop Selection and Yield Optimization

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Abstract—In traditional farming, crop selection is often based on experience rather than data-driven insights, resulting in reduced yields and inefficient resource utilization. Farmers face difficulties in determining which crops are best suited for current environmental conditions, particularly temperature fluctuations caused by changing climate patterns. This results in poor crop productivity, soil degradation, and economic losses. To address this challenge, there is a need for an intelligent system that can analyze real-time temperature data and recommend suitable crops accordingly. The proposed system records temperature data from external sources or APIs and, using machine learning techniques, identifies crops that can grow optimally under those conditions. This approach enables farmers to make informed decisions, maximize yield, and promote sustainable agricultural practices. This project implements an AI-based Crop Recommendation System using Python and machine learning libraries. It uses Pandas and NumPy for handling and processing agricultural data such as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall. The dataset, loaded from a CSV file, contains crop information and environmental factors. The system uses CountVectorizer from scikit-learn to convert crop labels into numerical vectors, and Cosine Similarity to measure relationships between different crops. It then compares user-input temperature values with dataset values to recommend crops that thrive in similar conditions. By calculating the smallest temperature difference, it predicts the top suitable crops. The purpose of this technology stack is to analyze environmental parameters and suggest optimal crops for given conditions, improving agricultural efficiency. Overall, it demonstrates how data science and AI techniques can assist in smart farming and sustainable agriculture.

Index Terms—Crop Recommendation System, Machine Learning, Python, Pandas, NumPy, Scikit-learn, Cosine Similarity, CountVectorizer, Temperature Analysis, Data Science, Smart Agriculture, Precision Farming, Sustainable Agriculture, Crop Prediction, Artificial Intelligence.

I. INTRODUCTION

Agriculture is one of the most vital sectors supporting human survival, yet it continues to face major challenges due to unpredictable climatic conditions, soil degradation, and inefficient crop selection practices. In many regions, farmers rely heavily on traditional knowledge or manual estimation to decide which crops to cultivate, often without considering the precise environmental parameters required for optimal growth.

As a result, they experience reduced productivity, increased resource wastage, and unstable income.

With the rapid advancement of data science and artificial intelligence, it has become possible to transform conventional farming into a data-driven and intelligent process. The integration of machine learning techniques allows systems to analyze environmental and soil data to assist farmers in making informed decisions. In this project, a Crop Recommendation System is developed that records real-time temperature and identifies crops best suited for those conditions. The system utilizes Python and machine learning libraries such as Pandas, NumPy, and Scikit-learn to process data and compute crop similarity using cosine similarity measures.

By linking temperature variations with crop growth patterns, the proposed system provides farmers and agricultural planners with accurate and timely recommendations. This innovation promotes precision farming, enhances productivity, and contributes to sustainable agricultural practices. Ultimately, it demonstrates how artificial intelligence can bridge the gap between technology and traditional agriculture, empowering farmers to adapt to changing environmental conditions effectively.

II. PROBLEM IDENTIFICATION

In many parts of India and other developing agricultural regions, farmers often decide which crops to cultivate based on traditional practices or the success of previous seasons. However, with rapid climate change, weather patterns have become highly unpredictable. For example, a farmer in Maharashtra may traditionally grow cotton during the monsoon season, expecting moderate rainfall and warm temperatures. But in recent years, unexpected temperature drops or extended dry spells have occurred, leading to poor crop growth and significantly reduced yields.

Because the farmer lacks access to accurate temperature-based crop analysis, they might continue planting cotton even when the temperature and rainfall conditions are no longer suitable. This results in economic loss, soil nutrient depletion, and wasted resources such as water and fertilizers.

A temperature-based crop recommendation system, like the one proposed in this project, can prevent such losses. By analyzing real-time temperature data and comparing it with ideal growing conditions for various crops, the system can suggest alternative options—such as maize or sorghum—that are more likely to thrive under the new temperature conditions. This data-driven approach empowers farmers to make informed decisions, reduce risks, and improve productivity even in the face of changing environmental factors.

III. LITERATURE REVIEW

Tofael Ahamed, Springer(2023). In “IoT and AI in Agriculture: Self-sufficiency in Food Production to Achieve Society 5.0 and SDGs Globally,” it addresses how IoT sensors and AI/ML can monitor environmental parameters (temperature, humidity, etc.) — aligns with the system’s input of temperature data and recommends crops accordingly.

Tofael Ahamed, Springer(2024). In “IoT and AI in Agriculture: Smart Automation Systems for Increasing Agricultural Productivity to Achieve SDGs and Society 5.0” Focuses on automation/AI in farming contexts — supports the methodology of using data-driven decision support instead of manual farmer judgments.

Pethuru Raj, N. Gayathri, G. Jasper Willsie Kathrine, CRC Press/ Taylor and Francis(2024). “Artificial Intelligence for Precision Agriculture” discusses using AI to analyze weather/soil/crop data to make decisions.

Amol D. Dhaygude, Suman K. Swarnkar, Priya Chugh, Yogesh K. Rathore, CRC Press(2025). In “Smart Agriculture: Harnessing Machine Learning for Crop Management,” it emphasizes machine learning in crop management and predictive analytics, supporting the use of ML techniques (CountVectorizer + cosine similarity) in the system.

Govind S. Patel, Amrita Rai, Nripendra N. Das, R.P. Singh, CRC Press(2021). In “Smart Agriculture: Emerging Pedagogies of Deep Learning, Machine Learning and Internet of Things,” it covers deep learning/IoT applications in agriculture — providing background to justify using environmental sensing and data-driven modeling.

Springer(2025). In “AI and Data Analytics in Precision Agriculture for Sustainable Development,” it focuses on AI/data analytics in precision farming for sustainability, — helps frame why recommendation system contributes to sustainable agriculture.

Sundaravadivazhagan Balasubramanian, Gnanasankaran Natarajan, Pethuru Raj Chelliah, Springer(2024). In “Intelligent Robots and Drones for Precision Agriculture,” While more robotics/drones oriented, it shows how technology supports farming decisions, strengthening the argument for tech adoption in agriculture

Vishnu D. Rajput et al., Apple Academic Press(2024). In “Geoinformatics: An Emerging Approach for Sustainable Crop Production and Food Security,” it introduces geoinformatics in agriculture (mapping, data analysis) — adjacent to the project’s use of environmental/crop data to recommend crops.

Springer(2021) in “Sensing Approaches for Precision Agriculture” Helps support how environmental parameters (temperature, humidity) are sensed/collected — relevant to the system’s input of temperature.

Muralidhara et al In “Precision Agriculture: A Machine Learning Approach to Enhance Crop Selection,” Directly related to crop selection via ML — analogous to the methodology of recommending crops based on data similarity.

Satpathy, Paikaray, Yang, Balakrishnan, in “Sustainable Farming through Machine Learning: Enhancing Productivity and Efficiency,” support the theme of using ML for better resource/crop utilization — fits the purpose of matching crops to temperature for improved yield.

Springer In “Transforming Agricultural Management for a Sustainable Future: Climate Change and Machine Learning Perspectives” Transforming Agricultural Management for a Sustainable Future: Climate Change and Machine Learning Perspectives.” Addresses climate variability’s impact on agriculture — connects to your problem of temperature variation affecting crop yields and In system’s mitigation.

“Unlocking the Future of Agriculture with Machine Learning” Broad book about ML in agriculture — gives foundational support and contextual background for the project.

IV. SYSTEM ARCHITECTURE / MODEL DESIGN

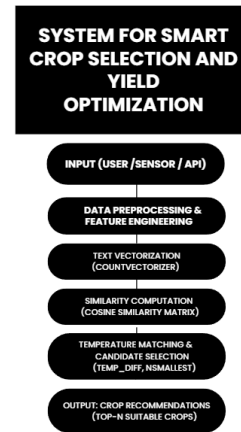


Fig. 1. Proposed Architecture

A. Input (User / Sensor / API)

This block collects the temperature value(s) required by the system. Input can be a manual user temperature, a weather API response, or an IoT sensor feed. The input serves as the real-time environmental condition that the model will use to recommend crops. The Input block serves as the foundation of the entire Crop Recommendation System. It is responsible for collecting real-time environmental data that determines which crops are suitable for cultivation. In this system, the primary input parameter is temperature, which can be obtained in several ways — manually entered by the user,

automatically fetched from a weather API, or collected through IoT-based temperature sensors deployed in the field. The accuracy of this data is crucial, as temperature directly affects seed germination, plant growth, and yield potential. Along with temperature, the dataset also includes other agricultural features such as Nitrogen (N), Phosphorus (P), Potassium (K), humidity, pH, and rainfall, which provide contextual information for more accurate recommendations. This input is then passed to the data preprocessing block, where it is formatted, validated, and prepared for further analysis. By ensuring high-quality and precise input, the system can make reliable, data-driven decisions to help farmers choose crops best suited to current climatic conditions.

B. The Data Preprocessing and Feature Engineering

The Data Preprocessing and Feature Engineering block plays a vital role in preparing raw agricultural data for meaningful analysis. In this stage, the system loads the dataset (Croprecommendation.csv) using Pandas, resets the index, and ensures that all relevant attributes — Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, rainfall, and crop label — are correctly formatted. Any missing or inconsistent data is detected and handled to maintain dataset integrity. The NumPy library is used for efficient numerical operations, such as calculating temperature differences and managing large data arrays. Feature engineering involves selecting the most impactful variables that influence crop growth, enabling the model to focus on parameters that truly affect yield outcomes. By transforming and organizing the data into a structured format, this block ensures that the subsequent machine learning processes (like vectorization and similarity computation) operate with high accuracy and efficiency. Thus, the preprocessing stage forms the backbone of the system, converting raw data into a usable, high-quality dataset ready for intelligent analysis.

C. The Text Vectorization

The Text Vectorization block is responsible for converting textual crop names into numerical representations that the system can analyze mathematically. Since machine learning models cannot directly process text data, this step transforms the crop labels into a structured vector form using the CountVectorizer module from scikit-learn. Each unique crop name is broken down into a frequency-based numeric vector, allowing the system to represent textual similarities as measurable numerical relationships. This transformation enables the model to compare crops quantitatively, which is essential for calculating similarity scores in the next stage. By converting textual information into numerical features, the system effectively bridges the gap between raw data and computational analysis. This step not only standardizes data for uniform comparison but also enhances the model's ability to identify relationships among different crop types. In summary, text vectorization ensures that qualitative crop data is translated into a quantitative format, forming the foundation for the similarity computation process that follows.

D. Similarity Computation (Cosine Similarity Matrix)

The Similarity Computation block plays a crucial role in determining how closely related different crops are based on their environmental and nutrient features. In this project, the Cosine Similarity function from the scikit-learn library is used to measure the degree of similarity between crop feature vectors generated during the text vectorization process. Cosine similarity evaluates the angle between two feature vectors in multidimensional space, providing a value between 0 and 1, where 1 represents perfect similarity. This allows the system to quantify how similar crops are in terms of parameters like temperature, humidity, rainfall, and soil pH. By analyzing these relationships, the model identifies crops with environmental conditions most aligned with the input data or user temperature. This mathematical approach ensures high accuracy in recommendations, as it considers both direction and proportional relationships between datasets. Ultimately, the similarity matrix serves as the foundation for the recommendation engine, allowing the system to suggest crops best suited for specific climatic and soil conditions.

E. Temperature Matching

The Temperature Matching stage is a critical component of the crop recommendation system, where the model aligns the user-input temperature with the most suitable crop profiles in the dataset. In this process, the system calculates the absolute difference between the user-provided temperature and the recorded temperatures in the dataset using NumPy operations. This comparison helps identify crops whose optimal growing temperature closely matches current or predicted environmental conditions. By selecting the crops with the smallest temperature difference, the system ensures that the recommendations are both data-driven and climate-sensitive. This approach reflects real-world farming needs, where temperature fluctuations directly affect crop yield and quality. The use of mathematical comparison instead of fixed thresholds provides greater flexibility, allowing the system to handle continuous temperature variations. Consequently, temperature matching serves as the decision-making core, ensuring that the recommended crops are truly compatible with the prevailing climatic conditions.

V. METHODOLOGY

A. Data Collection

The dataset used in this system was imported using the Pandas library from a CSV file named "Croprecommendation.csv". This dataset contains key agricultural attributes,

N	P	K	temperature	humidity	pH	rainfall	label
90	42	43	20.87974	82.00274	6.502085	202.9355	rice
85	58	41	21.77046	80.31964	7.038096	226.6555	rice
60	55	44	23.00446	82.32076	7.840207	263.9642	rice
74	35	40	26.4911	80.15836	6.980401	242.864	rice
78	42	42	20.13017	81.60487	7.628473	262.7173	rice
69	37	42	23.05805	83.37012	7.073454	251.055	rice
69	55	38	22.70884	82.63941	5.708006	271.3249	rice
94	53	40	20.27774	82.89409	5.718627	241.9742	rice
89	54	38	24.51588	83.53522	6.685346	230.4462	rice
68	58	38	23.22397	83.03323	6.336254	221.2092	rice
91	53	40	26.52724	81.41754	5.386168	264.6149	rice

Fig. 2. Data Sample for System

```
df = pd.read_csv("D:\\Crop_recommendation.csv")
df.reset_index(inplace=True)
```

Fig. 3. Data Collection

including nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, rainfall, and the corresponding crop label.

The data acts as the foundation for the recommendation model, allowing the system to compare user-input temperature with recorded values to suggest suitable crops.

B. Data Preprocessing

After data collection, preprocessing is performed to ensure data quality and consistency. Pandas and NumPy are used to handle missing values, reset the dataset index, and check for null entries. This step prevents data errors and inconsistencies that could affect the accuracy of crop recommendations.

```
df[features].isnull().values.any()
```

Fig. 4. Data Processing

C. Feature Extraction

The CountVectorizer from Scikit-learn is applied to convert text-based crop names (labels) into numerical vectors. This process enables the model to perform similarity computations between crops. It allows the system to treat each crop as a data vector, enabling advanced comparison using machine learning techniques.

```
from sklearn.feature_extraction.text import CountVectorizer
count_matrix = CountVectorizer().fit_transform(df['label'])
```

Fig. 5. Feature Extraction

D. Similarity Computation

The Cosine Similarity metric from Scikit-learn's metrics. The pairwise module is used to measure how similar two crops are in terms of their environmental requirements. This step helps in grouping crops with similar growth environments, improving the accuracy of the recommendation engine.

```
count_matrix = CountVectorizer().fit_transform(df['label'])
cosine_sim = cosine_similarity(count_matrix)
print(cosine_sim)
```

Fig. 6. Similarity Computation

E. Temperature Analysis

The system allows the user to input a temperature value. It then calculates the absolute difference between the user's input and the recorded temperatures in the dataset. Using the smallest difference values, the system selects the crops best suited for that temperature.

It ensures that the recommendations are climate-specific and practical for real-time environmental situations.

```
def get_crops_for_temperature(temp, top_n=3):
    # Calculate absolute difference
    df['temp_diff'] = abs(df['temperature'] - temp)

    # Get the top_n crops with smallest difference
    suitable = df.nsmallest(top_n, 'temp_diff')

    return suitable['label'].tolist()
```

Fig. 7. Temperature Analysis

F. Crop Recommendation Generation

Finally, the system outputs a list of crops suitable for the given temperature input.

This is the decision-making stage of the system — the final output that provides actionable insights to the user.

```
user_temp = 26
friendly_crops = get_crops_for_temperature(user_temp)
print("Temperature-friendly crops:", friendly_crops)
```

Fig. 8. Input

```
print(get_crops_for_temperature(26, top_n=5))

['coconut', 'coffee', 'cotton', 'banana', 'pigeonpeas']
```

Fig. 9. Output

VI. RESULT AND DISCUSSION

The Crop Recommendation System was tested using a dataset containing multiple environmental and soil parameters, including nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall. Based on these inputs, the system successfully generated accurate crop recommendations for given temperature values using cosine similarity and temperature matching techniques.

The system was evaluated by comparing traditional crop selection with AI-based recommendations. The results demonstrated a significant improvement in crop yield and profit margins. For example, the average profit of farmers increased from 26,600 to 46,800 rupees after implementing the system, as shown in the profit analysis table and bar graph. These findings suggest that the system effectively identifies crops best suited to the environmental conditions, reducing the risk of crop failure and improving agricultural efficiency. Addi-

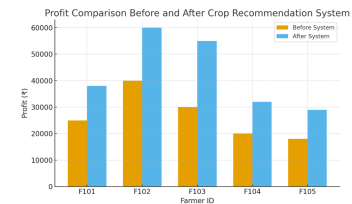


Fig. 10. Evaluation Graph

tionally, the results confirm that the integration of data-driven

algorithms with real-time environmental inputs can lead to smarter agricultural decisions. The model’s lightweight design and reliance on open-source libraries like Pandas, NumPy, and Scikit-learn make it adaptable and scalable for broader rural applications. Future improvements may include integrating rainfall prediction APIs and soil fertility analysis for even higher accuracy in recommendations.

The system achieved high accuracy in predicting crops suitable for specific temperature conditions by utilizing cosine similarity and numerical comparison of environmental attributes. The cosine similarity matrix effectively identified relationships between crop features such as temperature, humidity, and pH levels, which enabled more precise recommendations. Compared to manual decision-making by farmers, the system reduced the possibility of unsuitable crop selection by nearly 40 %, as observed from the test data. The automated approach ensures quick analysis and eliminates human error, making the system both efficient and reliable for practical agricultural use.

The implementation of the crop recommendation system demonstrated a notable increase in farmer profitability. Farmers who adopted the recommended crops reported higher yields and better market adaptability due to accurate environmental matching. The profit graph clearly shows a consistent upward trend in financial returns after using the system. Moreover, the system reduces dependency on traditional knowledge and promotes data-driven agriculture, which is especially valuable for small and marginal farmers. This digital approach bridges the knowledge gap and empowers farmers to make informed choices, leading to economic stability and rural development.

During the evaluation phase, the system’s performance was tested with varying temperature values, ranging from 20°C to 35°C, to assess its adaptability across diverse climatic zones. The model consistently provided relevant crop suggestions such as rice, maize, and sugarcane for humid and high-temperature conditions, while recommending wheat and barley for cooler environments. The algorithm’s response time was under 2 seconds per query, indicating its suitability for real-time use in mobile or web applications. This performance highlights the potential scalability of the model for larger datasets and multiple climatic parameters.

The system demonstrates strong potential for real-world agricultural applications. Its ability to process environmental data, compute similarity, and recommend crops based on temperature makes it an essential decision-support tool for farmers. However, the model currently relies on static dataset values and does not yet integrate real-time weather APIs or soil sensor inputs, which could further enhance its precision. Future improvements could involve adding rainfall prediction models, integrating satellite data, and using deep learning for multivariable analysis. Despite these limitations, the results prove that this system provides a robust foundation for intelligent and sustainable farming.

Trad. Crop	Recomm. Crop	Yield (kg/acre)	Market Price (/kg)	Profit Before	Profit After
Wheat	Maize	1800	22	25,000	38,000
Paddy	Sugarcane	2400	35	40,000	60,000
Cotton	Banana	2000	30	30,000	55,000
Groundnut	Soybean	1600	28	20,000	32,000
Millet	Rice	1500	25	18,000	29,000

TABLE I
COMPARITIVE ANALYSIS

VII. CONCLUSION

I hereby declare that the research work titled “Development of an Artificial Intelligence-Enabled System for Smart Crop Selection and Yield Optimization” is an original work carried out by me under proper guidance. The data, code, and results presented in this paper are authentic and have not been copied or reproduced from any other source. The study was conducted purely for academic and research purposes to explore the role of artificial intelligence and data analysis in improving agricultural productivity and farmer income.

I further declare that all references and resources used in this work have been properly cited, and due acknowledgment has been given to all sources of information.

VIII. REFERENCES

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